



±8Gauss, High Performance, Low Cost 3-axis Magnetic Sensor

MMC5883MA

FEATURES

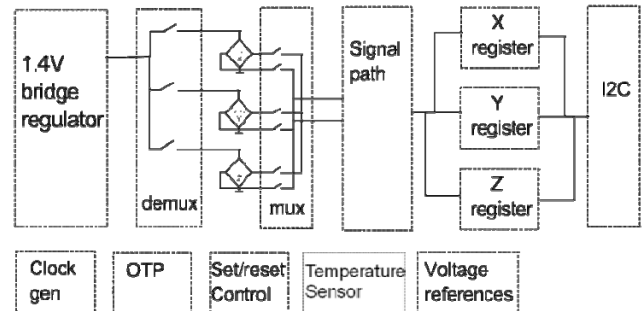
- Fully integrated 3-axis magnetic sensor and electronic circuits requiring fewer external components
- Superior Dynamic Range and Accuracy:
 - ✓ ±8G FSR with 16bits operation
 - ✓ 0.25mG per LSB resolution
 - ✓ 0.4mG total RMS noise
 - ✓ Enables heading accuracy of ±1°
- Max output data rate of 600Hz
- Industrial Standard Low profile package 3.0x3.0x1.0mm
- Degaussing with built-in SET/RESET function
 - ✓ Eliminates thermal variation induced offset error (Null field output)
 - ✓ Clears the residual magnetization resulting from strong external fields
- On-chip sensitivity compensation
- On-chip temperature sensor
- Motion_Detection and Data_Ready Interrupt
- Low power consumption
- 1μA power down current
- I²C Slave, FAST(≤400KHz) mode
- 2.5V single low power supply
- 1.8V I²C interface
- RoHS compliant
- P2P compatible with HMC5883L

APPLICATIONS

- Electronic Compass & Navigation
- Position Sensing
- General Purpose Magnetic Field Measurements

DESCRIPTION

The MMC5883MA is a complete 3-axis magnetic sensor with on-chip signal processing and integrated I²C bus suitable for use in various applications. The device can be connected directly to a microprocessor, eliminating the need for A/D converters or timing resources. It can measure magnetic fields within the full scale range of ±8Gauss (G), with 0.25mG per LSB resolution at 16bits operation mode and 0.4mG total



FUNCTIONAL BLOCK DIAGRAM

RMS noise level, enabling heading accuracy of ±1° in electronic compass applications. Contact MEMSIC for access to advanced calibration and tilt-compensation algorithms.

An integrated SET/RESET function provides for the elimination of error due to Null Field output change with temperature. Temperature information from the integrated temperature sensor is available over the I²C Interface. The SET/RESET function can be performed for each measurement, periodically, or when the temperature changes by a predetermined amount as the specific application requires.

In addition, the SET/RESET function clears the sensors of any residual magnetic polarization resulting from exposure to strong external magnets.

The MMC5883MA is packaged in a low profile LGA package (3.0 x 3.0 x 1.0 mm) and an operating temperature range from -40°C to +85°C.

The MMC5883MA provides an I²C digital output with 400 KHz, fast mode operation.

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